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Deep generative reconstruction of fibrous electrode architectures in PEM fuel cell electrodes

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ABSTRACT

Three-dimensional (3D) characterization of gas diffusion layers (GDLs) is vital for understanding transport phenomena and improving proton exchange membrane fuel cell (PEMFC) performance. However, most existing microstructure generation methods assume isotropy or periodicity, which fails to capture the anisotropic and heterogeneous nature of GDLs. In this work, we propose a conditional generative adversarial network (C-GAN) framework for reconstructing realistic multiphase fibrous microstructures from X-ray computed tomography (XCT) data. The model integrates fiber-aligned preprocessing and stochastic generation strategies to preserve anisotropic fiber orientation, binder distribution, and phase connectivity. Validation using TauFactor analysis shows close agreement with the structural characteristics of the original GDL, including tortuosity (1.99 vs. 1.85), phase volume fractions, and binder-fiber interfacial area (~2.8% deviation). With a generation time of approximately 35 min, the framework offers an efficient and scalable alternative to experimental imaging, enabling digital twin development and structure-property studies in hydrogen energy systems.

1. Introduction

The growing global energy demand historically encountered via the combustion of traditional fossil fuels has led to substantial environmental pollution and accelerated climate change. In this regard, the development and implementation of alternative, environmentally sustainable energy sources have become essential to ensure a cleaner and more resilient energy perspective. Renewable energy sources, including solar, wind, hydro, geothermal, and biomass, have gained considerable attention, although their large-scale implementation is limited by several key factors: intermittent power generation, limited energy storage capacity, high production costs, and scarcity of materials [1]. Additionally, challenges faced in grid integration and system reliability further limit operational efficiency. Hydrogen has been regarded as a promising clean energy carrier with high energy density and strong potential for decarbonization [2]. Traditional technologies for hydrogen production utilizing natural gas and coal that are used in steam reforming substantially contribute to CO_2 emissions. Unlike alternative energy technologies and traditional hydrogen-producing technologies, fuel cells are capable of continuous and controllable power generation, high energy efficiency, and reliable operation, making them better-suited candidates for sustainable energy generation on a global scale. Specifically, hydrogen fuel cells that electrochemically combine hydrogen and oxygen from air to generate electricity are considered efficient energy converting devices with zero greenhouse gas emissions.

Proton exchange membrane fuel cells (PEMFC) are promising electrochemical devices compatible with dynamic current loads, well adapted to operate in both stationary power applications and green hydrogen generation [3]. They possess several advantages, such as low operating temperature, rapid start-up and achieving high power density. They also demonstrate compatibility with fluctuating electrical demands. Despite these attractive properties, their large-scale

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commercialization is still limited by operational challenges concerning power density and efficiency, as well as the high costs of their main compartments. Cold-start performance, closely linked to water management and microstructural transport limitations, is another key challenge hindering PEM fuel cell commercialization [4].

A PEMFC consists of several key components: the catalyst layer, PEM, GDL conventionally made from carbon fiber paper (responsible for reactant gases distribution to the catalyst layer and removal of excess water generated *in operando*), and bipolar plates (responsible for distribution of gases, current collection, and maintaining the structural integrity of the fuel cell stack). The schematic of the working principle of PEMFC is shown in Fig. 1.

The capability of PEM fuel cells critically depend on their microstructural architecture and on their porous electrodes. These electrodes consist of multi-phase composites: a solid catalyst network, an ionomer binder domain, and a percolated gas-filled pore phase. Their microstructure governs charge-transfer kinetics, reactant diffusion, and thus ultimately determines mass-transport efficiency, power density, and durability of the PEM fuel cells. Nevertheless, despite the critical role of microstructural morphology, a comprehensive quantitative understanding of how it governs the macroscopic behavior of the PEM fuel cell is still lacking. This gap results from the challenges of obtaining and processing high-resolution, 3D imaging data capable of resolving all phases with statistically representative volumes.

State-of-the-art advanced imaging techniques, i.e., focused ion beam–scanning electron microscopy (FIB-SEM), X-ray computed tomography (XCT), and time-of-flight secondary ion mass spectrometry (ToF-SIMS), have substantially improved the structural characterization and provided unprecedented insights into electrode architecture and degradation phenomena [5–7]. However, each technique involves intrinsic trade-offs between achievable resolution, chemical sensitivity, and field of view. For example, XCT enables non-destructive 3D visualization over relatively large volumes but provides insufficient contrast to clearly distinguish low-atomic-number constituents, i.e., the carbon binder and pore domains [8]. In contrast, FIB-SEM achieves nanometer-level detail yet is limited to small sample regions and relies on destructive serial sectioning. Meanwhile, ToF-SIMS provides excellent chemical specificity and, when integrated with machine-learning algorithms, can resolve both major phases and interphases within electrodes [5]; however, extending this approach to statistically representative 3D volumes remains difficult. As a result, computational reconstruction techniques are increasingly adopted to augment or, in some cases, replace experimental imaging data.

The common approaches to PEM fuel cell GDL structure optimization were traditionally based on simulation techniques, including computational fluid dynamics (CFD), lattice Boltzmann methods (LBM), and pore network modeling (PNM). Despite their strong physical foundations, these techniques require substantial computational resources and are not well-equipped to handle broad design spaces or the stochastic variation common for porous materials. Recent study by Wang et al. has highlighted that even state-of-the-art large-scale simulations demand supercomputing resources and remain constrained by imaging and modeling limitations, underscoring the need for alternative approaches [9].

There has been rapid progress in this field over the past few years by utilizing machine learning (ML) tools. Deep learning architectures such as convolutional neural networks (CNNs) have shown a strong ability to learn complex relationships between porous microstructure and transport properties without relying on explicitly defined physical equations [10]. Such data-driven models have been utilized for the estimation of permeability, power output from raw tomographic data, as well as to reconstruct microstructures from incomplete data [11]. An optimization of the GDL microstructure in PEMFCs has been recently developed by Omongos et al. [12] yielding an R^2 value above 0.95 and showing negligible deviation between model predictions and measured results. The design and optimization of electrodes in PEMFCs involve tackling complex relationships between the microstructural characteristics of a multilayered architecture (carbon fiber paper, microporous layer, catalyst layer, GDL) and electrochemical performance.

Traditional experimental methods, often reliant on device fabrication and testing, are resource-intensive and lack scalability. This study hypothesizes that a computational approach leveraging digital twin technology, synthetic training data generation, and advanced deep learning algorithms can replicate and optimize these relationships. Recent advances in deep learning have significantly improved material imaging and segmentation [13,14], particularly for complex systems

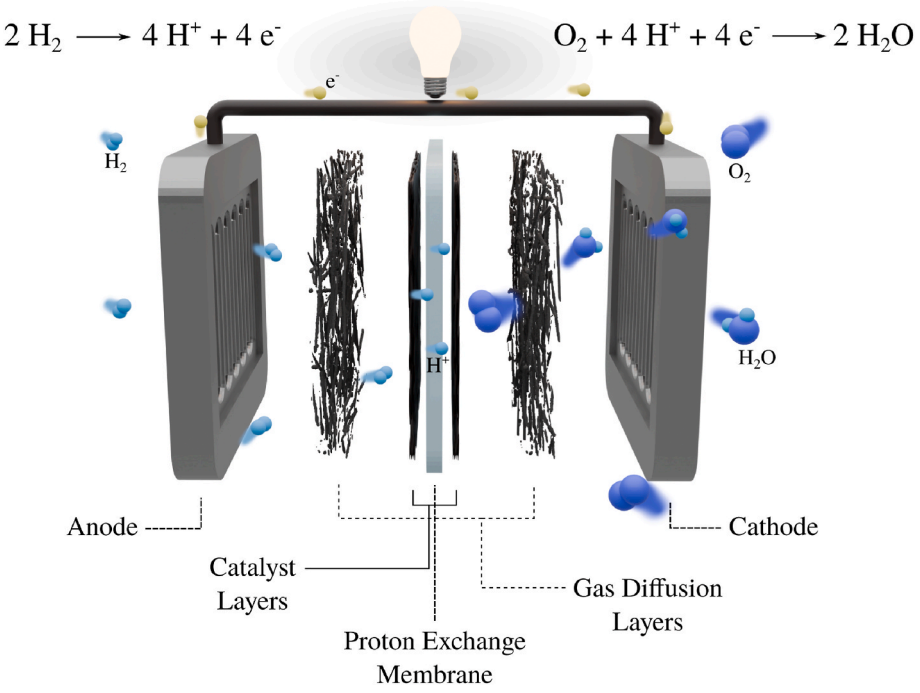


Fig. 1. Schematic of a proton-exchange membrane fuel cell.

such as fuel cells and batteries, i.e., several recent studies have explored generative models to synthesize or reconstruct 3D microstructures for electrochemical porous electrodes. Mosser et al. [15] pioneered the use of generative adversarial networks (GANs) for 3D reconstruction of two-phase porous media, with Lombardo et al. [16] extending this to multiphase electrode materials such as SOFC anodes and Li-ion cathodes. Notably, their architecture enabled them to produce periodic microstructures in all directions by adjusting the latent input vector. SliceGAN proposed by Kench and Cooper [17], further enabled 3D reconstruction by training on 2D slices, allowing flexible volumetric synthesis with tunable resolution. This approach has since been enhanced using conditional GANs (cGANs) for inpainting and outpainting tasks. These models generate large representative domains from limited data while preserving key statistics such as porosity, surface area, and connectivity.

Recent work by Kishimoto et al. [18] demonstrated that generative adversarial networks can successfully reconstruct porous microstructures with high morphological fidelity, using physical descriptors to guide training. Xu et al. [19] developed a conditional GAN framework for reconstructing solid oxide fuel cell anode structures under varying fabrication conditions, incorporating physical constraints such as particle size and porosity. Their model achieved strong agreement with experimental data and highlighted the potential of using GAN-based models for guiding electrode design. However, the reconstructions are limited to relatively isotropic structures and cannot recover the full anisotropic or layered morphology found in fuel cell GDLs. Sugiura et al. [20] proposed Big-Volume SliceGAN to mitigate training limitations on small volumes, enabling the synthesis of large representative domains. Their work demonstrated that increasing representative volume and resolution enhances structural continuity, although the method struggles with multi-phase complexity. Although conventional 3D GAN implementations are computationally intensive and often impractical for large-scale volumetric imaging. To overcome these limitations, Tang et al. [21] introduced P3T-Net, a pseudo-3D domain transfer that integrates CycleGAN principles with semantic consistency modules to enable efficient and scalable domain alignment for 3D material imaging. This approach significantly reduces memory requirements, mitigates third-axis misalignment, and preserves critical physical features across domains, thereby facilitating robust segmentation and analysis under diverse imaging conditions in geoscience, energy storage, and fuel cell applications.

These developments highlight the growing importance of deep learning and GAN-based models particular in overcoming imaging and processing challenges for electrochemical energy systems, a promising balance between morphological realism, speed, and data efficiency, making them suitable for microstructural modeling in hydrogen energy applications. However, while powerful, most of the existing GAN models emphasize relatively isotropic systems; in contrast, PEMFC GDLs pose additional challenges due to their anisotropy and multi-phase hierarchy. Although GAN frameworks have progressed significantly, they were originally designed for single-phase, statistically isotropic images, and extending them to fully capture 3D, anisotropic, and multi-phase microstructures remains challenging. Consequently, the layered architecture and strongly directional anisotropy that are intrinsic to PEMFC GDLs - arising from their fiber-oriented, multi-phase, and through-plane transport-dominated structure - remain generally underrepresented in existing GAN-based reconstruction methods, this gap persists particularly in fuel-cell specific literature.

To address this gap, this study aims to develop and validate a novel anisotropy-aware conditional generative adversarial network (C-GAN) framework for reconstructing multiphase fibrous GDL microstructures from X-ray computed tomography (X-ray CT) data. The framework includes fiber-aligned preprocessing to improve through-plane continuity and to mitigate the limited ability of 2D convolutional layers to capture long-range spatial features. It also accounts for anisotropy and multiphase structure, which are critical for transport processes such as gas diffusion, water management, and effective conductivity.

To evaluate the morphological fidelity and transport-relevant behavior, the generated microstructures are quantitatively assessed using TauFactor 2 tool [22]. This tool provides metrics including tortuosity, specific surface area, and phase volume fractions.

2. Methodology

2.1. Materials

The GDL used in this study is a widely used commercial material (SGL 29 BC, Sigracet, US). According to the manufacturer's specifications, this GDL is composed of carbon fiber, processed as carbon fiber web using papermaking technology and subsequent thermobonding [23]. This carbon fiber paper is then impregnated with a binder composed of resins cured and recarbonized/graphitized. The MPL is then deposited by the manufacturer and uses a 77 wt% carbon black and 23 wt% PTFE mixture. The morphology of the GDL was characterized using micro-CT. A laboratory micro-CT system (Mark I HeliscanTM microCT, ThermoFisher, USA) was used for 3D imaging. A cylindrical sample of 2 mm was extracted from the 1 cm^2 membrane electrode assembly using a biopsy punch to optimize the image collection process and maximize the signal-to-noise ratio. A source voltage and current of 60 kV and 190 μA were used. The sample was rotated over 360° axis, collecting 2880 projections, with an exposure time of 0.6 s and 5 accumulations per projection. The sample was placed 2.2 mm from the source, which resulted in an image pixel resolution of 900 nm. The data was originally captured by Tang et al. [13] provided for this study as training data.

2.2. Conditional GAN

GAN is a deep learning model that generates artificial datasets that are statistically and morphologically representative of actual structures. Despite their success, generating images of fuel cells with continuous and unique features remains a challenge. Existing models try to capture the main features; but they often fail to maintain consistency along the depth dimension (z-axis), which represents the progression of slices through the material. This lack of continuity leads to misalignment and inconsistencies in their 2D and 3D structures.

To overcome these limitations, this work employs a Conditional Generative Adversarial Network (C-GAN). The proposed approach follows the conditional GAN architecture, which consists of two competing networks: the generator and the discriminator. The following section provides a detailed description of these components.

The methodology is structured into two distinct coupled stages. First, the model learns the structure of the fuel cells, the chaotic behavior of carbon fibers, assigning orientation, and distribution patterns. In the second stage, slices are gradually refined and aligned with real data to enhance morphological accuracy. This combined strategy ensures continuity across slices while preserving physical properties derived from the original dataset.

There are six distinct phases of the GDL: Void, Catalyst, Membrane, Microporous Layer (MPL), Binder, and Carbon Fiber. These phases were categorized into strict pixel categories to better capture their physical behavior (see Table 1).

Probabilistic modeling was used to predict pixel values, enabling

Table 1
Key details about the training dataset.

Property	Training
Materials	PEMFC GDL (XCT) - SGL 29 BC, Sigracet, US
Dimensions/voxels	656 × 656 × 81
Cubic voxel size/nm^3	900 × 900 × 900
Number of subvolumes	1512 (128 × 128 × 81 voxels each)

smooth transitions and continuous structural evolution across slices. Table 2 summarizes the key characteristics of the training dataset used for model development. One-hot encoding was applied to enforce strict class distribution, as binary classification (1/0) for each of the six phases is more effective than raw probability values. If a pixel exhibits a high probability of belonging to a specific phase, it is classified accordingly, allowing the model to learn structural patterns. Images were constrained to pixel dimensions that are multiples of 16 to capture fine-scale features and expand them into larger domains. Padding was applied to prevent empty spaces and eliminate black artifacts in the final images.

2.2.1. Training strategy

One of the primary challenges was the lack of true comparison between real data and generated data during training, making it difficult for the model to recognize when improvements were occurring rather than producing random pixel variations. To mitigate this, controlled noise was added to the original slices, enabling the model to learn how to reduce noise and generate progressively improved outputs at each training step. A variable *input_idx* represents the noisy image, while *real_idx* corresponds to the original slice. When *input_idx* becomes closer to the real image, the model is rewarded, to reinforce the improvement cycle. The training was done on PyCharm v.2024.3.5 with PyTorch v2.9.0.

During training, synthetic input-target pairs $(y,\widehat{y})$ were dynamically generated. For a given ground-truth slice (y), we create a corrupted input $(\widehat{y})$ via two operations:

Spatial Shifting: Fiber channels were spatially randomized to prevent the model from memorizing exact locations. It was necessary to use a 2D grid for randomizing those locations by applying on the pixel map. This grid is called a tensor, a matrix of numbers that will determine the locations of the specific colors that were deduced from the classes. This was achieved using *torch.roll* with random offsets (Δx,Δy) ensuring spatial variability and reducing rigidity in fiber placement. *Torch* is a library from PyTorch that changes the matrix of the pixel map with the function "*roll*".

Noise Masking: The binder phase in the original dataset exhibits an organic, evolving structure with pores, requiring smooth transitions rather than abrupt flickering. To replicate its foamy morphology, a parameter, N_{foam}, was introduced, along with constraints on porosity to maintain realistic tortuosity levels $\tau \sim U(0.35,0.65)$. These constraints prevent uncontrolled randomness and preserve structural integrity during generation.

Table 2
Refined generator architecture.

Block	Layer Type	In	Out	Normalization	Activation
Encoder					
E1	Conv2d (44), stride 2	C = 6	64	-	LeakyReLU
E2	Conv2d (44), stride 2	64	128	BatchNorm	LeakyReLU
E3	Conv2d (44), stride 2	128	256	BatchNorm	LeakyReLU
E4	Conv2d (44), stride 2	256	512	BatchNorm	LeakyReLU
Bottleneck					
B1	Concatenate z ~ N(0, I)	512	512 + 64	-	-
B2	Conv2d (3 × 3), stride 1	576	512	-	ReLU
Decoder					
D1	Upsample + Conv2d + Concat(E3)	512	512	BatchNorm	ReLU
D2	Upsample + Conv2d + Concat(E2)	512	256	BatchNorm	ReLU
D3	Upsample + Conv2d + Concat(E1)	256	128	BatchNorm	ReLU
D4	Upsample + Conv2d	128	32	-	ReLU
Output					
Out	Conv2d (3x3, stride 1)	32	C = 6	-	Linear (Logits)

2.2.2. Architecture

To generate images for the discriminator, we employed a Stochastic U-Net Generator. The function of the encoder is to compress the input image to learn structural patterns and then the decoder reconstructs the image based on these learned features. Since a standard U-Net produces deterministic outputs, a noise variable z was introduced to incorporate randomness. This stochastic approach gives a natural variability and smoothness to a binder. Since the image is shrunken by the encoder, fine details, particularly binder edges, can be lost; therefore, skip connections were implemented to preserve critical features and prevent excessive blurring. These connections act as bridges between encoder and decoder, maintaining fiber integrity, which is visually prominent compared to other phases. In practice, this is achieved using concatenation operations, for example:

```
b = torch.cat([e4, z_tiled], dim = 1),
```

where the compressed representation *e4* is combined with the tiled noise vector z. This mechanism provides structural guidance while introducing controlled stochasticity, enabling the generator to produce realistic and diverse microstructures.

The Discriminator plays a critical role in determining whether the generated image sufficiently approximates the properties of the original. Since the model handles multiple phases, evaluation must go beyond a simple pixel-wise true/false classification and instead assess structural consistency across classes. To achieve this, a PatchGAN architecture was employed (Table 3), which operates on local image patches rather than the entire image, enabling fine-grained analysis of morphological features.

For example, a convolutional layer such as *nn.Conv2d(256, 1, 4, 1, 0)* aggregates information across all classes rather than producing a single binary output. The discriminator does not reject an image for minor deviations, as doing so could destabilize training and prevent convergence. Instead, it evaluates local regions using a 16 × 16 patch-based approach applied across the generated image, as implemented in PatchGAN. This method enables the discriminator to assess whether fibers exhibit realistic spatial behavior rather than relying solely on global visual similarity. Fibers, being the most complex and visually prominent structures, are analyzed through sharp intensity gradients - key indicators of linear features. By applying this patch-based grid, the discriminator verifies whether fiber continuity and orientation align with the original 3D structure. To avoid abrupt discontinuities between slices, a normalization constraint analogous to a 'speed limit' is introduced, penalizing excessive changes and guiding the generator toward smooth transitions. These constraints are represented in a matrix-like grid, where differences are computed and used for feedback. Through this iterative interplay between generator and discriminator, the model progressively learns to produce images that preserve structural integrity and physical realism.

Calculating the loss function is essential to monitor whether training is progressing in the right direction. In this work, we use Hinge Loss, which offers the advantage of enforcing a safety margin between real and generated samples. Real images are assigned a target of +1, while fake images are assigned −1. The discriminator loss is computed as follows:

Table 3
Discriminator architecture (D).

Layer	Layer Type	In	Out	Normalization	Activation
L1	Conv2d (44), stride 2	C = 6	64	Spectral Norm	LeakyReLU
L2	Conv2d (44), stride 2	64	128	Spectral Norm, Instance Norm	LeakyReLU
L3	Conv2d (44), stride 2	128	256	Spectral Norm, Instance Norm	LeakyReLU
L4	Conv2d (44), stride 1	256	1	Spectral Norm	Linear

$$Loss_D = \frac{1}{N}\sum \left(ReLU(1 - d_{real}) + ReLU(1 + d_{fake})\right)$$

where $Loss_D$ is discriminator loss;

ReLU (Rectified Linear Unit): When the prediction is sufficiently accurate, the loss approaches zero. If inaccuracies remain, the loss reflects the magnitude of deviation from the original, providing a clear signal for corrective updates during training;

d_{real}**:** Score for real microscopic data

d_{fake}**:** Score for fake data.

Hinge Loss operates by penalizing predictions that fall within a margin, using the *ReLU* function to ensure non-negative values. For real samples, the term $ReLU(1 - d_{real})$, is computed. For example, if a real image receives a score of 1.5, the calculation becomes 1.0 - 1.5 = −0.5, which *ReLU* maps to zero, indicating no penalty since the prediction exceeds the margin. Similarly, for fake samples, the term $ReLU(1 + d_{fake})$ is used. This pushes fake predictions toward −1, encouraging the discriminator to separate real and generated data while maintaining geometric stability.

Unlike binary classification losses, this approach avoids harsh penalties for minor deviations, reducing instability and mode collapse. The generator loss complements this by reversing the objective: it seeks to maximize the discriminator's confidence in generated samples. To enforce structural fidelity, we incorporate a Cross-Entropy Content Loss, which computes pixel-wise classification accuracy across all phases, with particular emphasis on carbon fibers and binder regions. The ground truth in this context is the original segmentation dataset, and the content refers to objects and shapes found in the slices of the said dataset. The formula is given below

$$Loss_G = \lambda_{ADV} \cdot \left(-\frac{1}{N}\sum d_{fake}\right) + \lambda_{CE} \cdot Loss_{CE}$$

where λ_{ADV}: The weighting of visual realism.

λ_{CE}:The weighting that enforces strict adherence to the material blueprint.

d_{fake}**:** The score assigned by the judge to images.

$Loss_{CE}$: The Cross-Entropy error that ensures each pixel is assigned the correct material category.

The generator produces a structural blueprint based on learned patterns, while the discriminator evaluates its fidelity to real microstructures. The discriminator is responsible for assessing the generated blueprint. The generator is trying to get the highest possible score from discriminator via d_{fake}. Weighted loss components (λ_{ADV} and λ_{CE}) are introduced to balance adversarial learning with pixel-wise accuracy, ensuring proper placement of fibers and binder so that physical properties are preserved. If the discriminator score falls below a threshold, penalties are applied to guide the generator toward improvement. These weightings act as conditional variables that enforce structural rules: adversarial error should remain low, while content-based error should be minimized for critical phases. The full training pipeline is illustrated in Fig. 2.

After the training, the most critical step is the generation process, which must maintain a continuity along the z axis. It starts with kinematic simulation. First, carbon fibers are modeled as 2D rigid bodies. Fibers were modeled by first identifying their corresponding pixel class and extracting relevant properties. Each fiber was assigned an orientation to reflect its physical behavior, ensuring continuity as a rigid entity traversing the domain. Position updates across successive slices were computed using a velocity vector evolving under random acceleration and drag forces ($P_{t+!} = P_t + V_t$). To maintain physical realism, directional changes were constrained to small incremental steps, avoiding abrupt rotations or discontinuities. Similarly, acceleration was bounded to prevent unrealistic fluctuations in motion. This controlled stochastic evolution preserves smooth trajectories and ensures that fibers maintain continuous, cylindrical paths throughout the reconstructed volume. Secondly, a provisional binder mask is generated using distance-gated noise. A “foam” noise field is thresholded to create potential binder regions, but these are strictly gated by a distance transform derived from the fiber positions. This encodes the physical prior that the binder must adhere to the carbon scaffold*:*

$$Mask_(Permit) = \{\, p \in \Omega |\ dist(p, Fiber) \le \tau_(dist) \,\}$$

where $Mask_(Permit)$ is the binary permission zone where the binder phase is mathematically allowed to exist.

p: An individual pixel coordinate within the two-dimensional image grid;

Ω: The total spatial domain representing the entire image area;

$dist(p, Fiber)$*:* The Euclidean distance calculated from pixel p to the

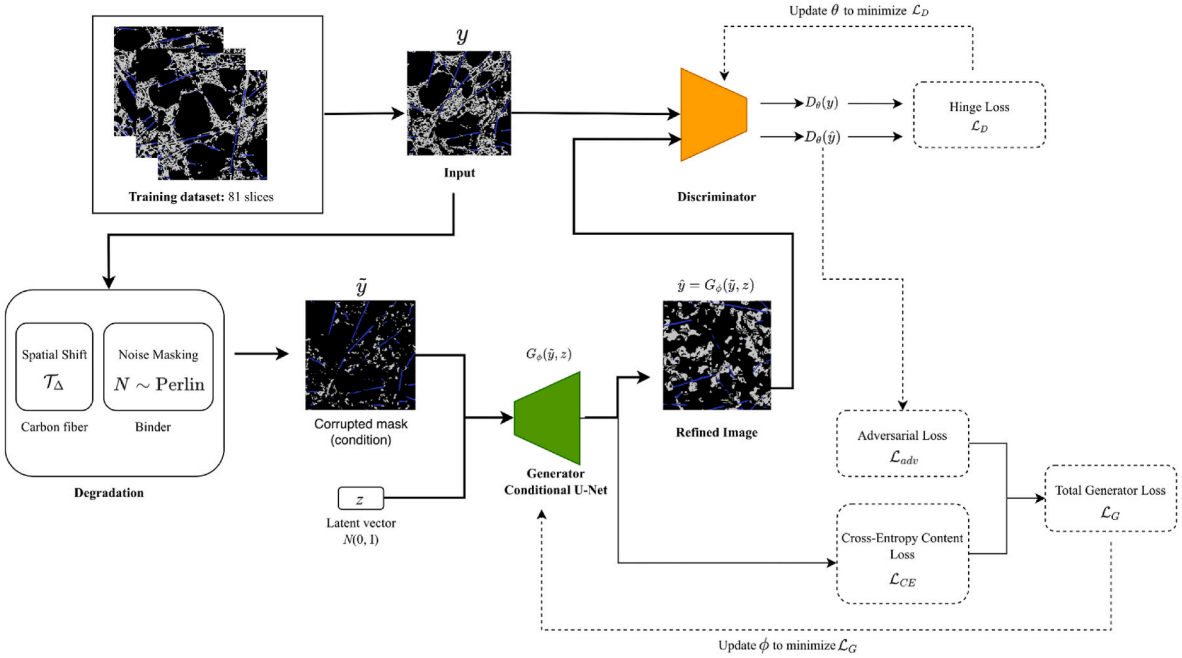


Fig. 2. GAN Architecture with full training process.

boundary of the nearest carbon fiber;

$\tau_(dist)$: The predefined distance threshold (25 pixels) that constrains the interaction between binder and fiber.

Finally, the coarse structural proposal is refined by the trained Generator. The network reconstructs realistic surface textures, smooths binder - fiber interfaces, and eliminates non-physical artifacts. To mitigate minor semantic leakage - specifically the erroneous prediction of microporous layer (MPL) phases within void regions - a post-processing routine is applied, consisting of Gaussian smoothing (σ=*1.5)* followed by morphological opening to remove disconnected artifacts smaller than 60 pixels. This combined approach yields a chemically consistent and structurally continuous three-dimensional volume.

The experiments were conducted using an NVIDIA RTX 5080 GPU with 16 GB of video memory, supported by 32 GB of system RAM and an AMD Ryzen 7 9700X 8-core processor. The training and image generation process required approximately 35 min to complete.

2.3. Quantitative analysis

To validate the synthetic microstructures, a quantitative analysis was performed using the following metrics: tortuosity along the z-axis, phase volume fractions, specific surface area of individual phases, and interfacial area between binder and carbon fiber phases. All calculations were performed using TauFactor [22], an open-source toolbox for analyzing transport properties, surface metrics, and porous media. The tortuosity values were calculated using the two-phase solver (conducting and non-conducting phases), and the "gradient" method was used in the calculation of the specific surface areas from the voxelized structures.

For each phase iii, the specific surface area is the interfacial area per unit volume,

$$S_{\mathrm{V,i}} = \frac{S_{A,i}}{V} = \frac{1}{V}\int dS_{A,i},$$

where $S_{A,i}$ is the total interfacial area between phase i and all other phases, and V is the total domain volume.

On voxel data, $S_{A,i}$ is estimated by TauFactor using a morphological-gradient/neighbor-count evaluation over a $3 \times 3 \times 3$ stencil that identifies surface voxels and accumulates their exposed area. Relative specific surface areas were reported by normalizing to the void-phase surface area $S_{\mathrm{V,void}}$ which is typically the largest and thus a stable reference.

We use the transport-based definition of the tortuosity factor τ, relating the effective diffusivity D_{eff} to the intrinsic diffusivity D_0 as:

$$D_{eff} = D_0 \frac{\varepsilon_i}{\tau}$$

where ε_i is the volume fraction of the conducting phase. Equivalently, τ follows from the ratio of steady-state flux through the porous phase to that through an equally sized fully dense control volume under the same driving gradient:

$$F_p = A_{cv} D_0 \frac{\varepsilon_i}{\tau}\frac{\Delta C}{L_{CV}}, F_{CV} = A_{CV} D_0 \frac{\Delta C}{L_{CV}}, \Rightarrow \tau = \varepsilon_i \frac{F_{CV}}{F_p}$$

TauFactor computes F_p by solving the steady diffusion equation on the voxel grid in two-phase mode (conducting vs. non-conducting). The Dirichlet boundaries along z (fixed concentrations at inlet/outlet) and no-flux (Neumann) boundaries on the x–y faces to obtain the through-plane tortuosity τ_z.

Bulk properties were computed using the entire volume of each dataset. For the relatively specific surface area normalization, the surface area of the void was chosen as the reference due to its highest surface area, ensuring consistency across evaluations. To estimate the binder–carbon fiber interfacial area, the datasets were remapped so that all phases except the binder and the carbon fiber turned into void. The fact that the "void" phase now contains several solid phases is not critical, as this is done only as a means of calculating the interfacial area between the binder and the carbon fiber phases. After the remapping, the specific surface areas (S_i) of the existing phases were calculated, and the interfacial area was estimated according to the following equation:

$$A = \frac{S_{binder} + S_{fiber} - S_{void}}{2},$$

where A is the relative interfacial area.

For the relative interfacial area, the surface area of the joined "void" phase is taken as a reference.

To capture spatial heterogeneity, 300 random subvolumes of the dimensions $100 \times 100 \times 25$ voxels were selected from each full dataset. The metrics were calculated in the same manner as for the bulk properties, except for the relative surface areas; instead of using each subvolume's surface area of the void phase as a reference, the mean surface area of the void phase across all subvolumes was used.

3. Results and discussion

A non-woven GDL (SGL 22 BB, Sigracet) was used, which is a common material for hydrogen fuel cell research [24–26]. Throughout this study, multiple 3D GAN architectures proposed in the literature were reviewed and evaluated for their applicability to reconstructing the microstructure of GDLs. SliceGAN, a patch-based generative adversarial network optimized with independent discriminators from 3 orthogonal planes, was tested to reconstruct the GDL during this work. While SliceGAN has demonstrated success in reconstructing stochastic porous media such as sandstone or battery cathodes, in GDL reconstruction, we faced a fundamental limitation. The receptive field of the model is limited and could not capture the full length of the carbon fiber. Given the computational constraints, the receptive field was limited to 128^3 voxels, and the model could validate that a structure appears cylindrical locally but lacked the global context to enforce Euclidean straightness across the entire volume. To overcome volume constraints, cube-stitching and tiling techniques were explored to generate volumes with 256^3 and 512^3 voxels, the generated volumes failed to maintain global geometric consistency in the sample. This indicates that patch-based convolutional models such as SliceGAN are not suitable for microstructure generation with both directional and layered anisotropy.

Figs. 3 and 4 allow for a visual comparison of the training and generated datasets. In Fig. 3 the presence of isolated catalyst-class voxels within the GDL originates from the raw segmentation rather than physical deposits; these voxels account for <0.1% of volume and are removed during semantic post-processing. The XCT training images clearly exhibit anisotropic layering with fibrous carbon structures embedded within a porous binder-MPL matrix. The synthetic 3D reconstructions mimic this complex, non-periodic morphology, including fiber alignment, interfacial continuity, and multi-phase embedding. Importantly, the GAN-generated microstructures maintain the anisotropic and layered morphology, preserving the orientation and connectivity of carbon fibers while capturing the random binder material distribution. The model generates structurally unique outputs each time, highlighting its capacity for stochastic variability while maintaining key morphological features, which is further supported by the analysis conducted on two more independent reconstructions, as detailed in the Supplementary Information.

Fig. 5 demonstrates the three dimensional representation of carbon fibers within the generated microstructure viewed from two perspectives and illustrates continuous fiber trajectories through successive z-slices, validating the volumetric representation of the generated reconstructions. This also demonstrates the ability of the model to capture extended morphological consistency, which is crucial for the modeling of anisotropic fibrous networks in GDLs.

Table 4 summarizes the comparison of bulk morphological metrics

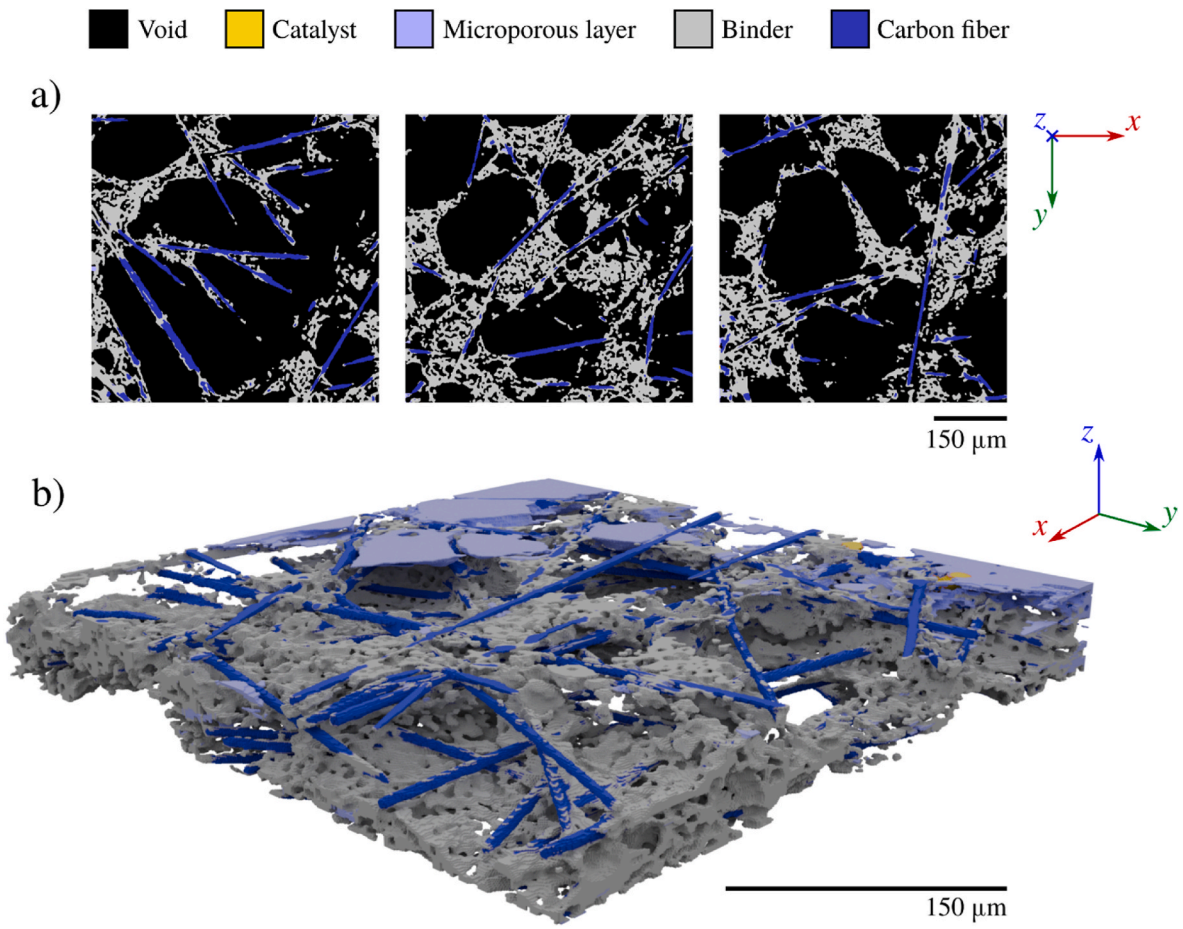


Fig. 3. Representation of the training dataset. a) Segmented X-Ray CT scans; b) 3D representation of the scans.

between the training and generated volumes. Tortuosity values are closely matched (1.99 vs. 1.85), indicating that the generated microstructures replicate pore-phase connectivity along the through-plane axis with ~7% deviation. This suggests that the model effectively learns the dominant transport pathways despite some local morphological variations. Such deviations reflect the systems' anisotropic transport behavior and subtle shifts in binder phase distribution or fiber arrangement.

The volume fractions across void, binder, and fiber phases remain consistent, with the largest deviation seen in the carbon fiber phase (−0.8%), attributable to minor truncation at volume boundaries. This affirms that the model reliably encodes and reconstructs phase distributions.

Specific surface area values show a modest reduction in the generated dataset, particularly for the void and binder phases. This trend is expected due to the smoothing effect of convolutional layers, which tend to suppress high-frequency surface roughness. While this may slightly reduce apparent surface complexity, the impact on electrochemical performance is likely minimal because the interfacial area between binder and carbon fiber, where most electrochemical reactions occur, is well preserved. Indeed, the binder–fiber interfacial area differs by only ~2.8%, and the relative interfacial metric increases slightly (0.089 to 0.105), suggesting that the GAN compensates for smoothing by maintaining connectivity at critical phase junctions.

Relative surface area ratios further confirm morphological fidelity, with binder and carbon fiber phases exhibiting trends consistent with the original dataset. These ratios are essential for modeling capillary transport and wetting behavior, and their preservation underscores the robustness of the proposed approach.

However, the interfacial area between binder and carbon fiber is well preserved, with only a ~2.8% difference, and a small increase in the relative interfacial metric (from 0.089 to 0.105), confirming morphological continuity at phase junctions. These results indicate that the model preserves both the volume-averaged structural properties and phase interface topology that are essential to PEM fuel cell GDL performance.

Fig. 6 presents the distributions of tortuosity, specific surface area, and interfacial area computed from 300 random subvolumes (100 × 100 × 25 voxels) for both real and generated datasets. Generated samples closely match the means of the training data across all metrics, with a narrower spread for tortuosity and binder-related surface measures, consistent with mild interface smoothing in convolutional generative models. Phase volume fraction distributions (Fig. 6b) align well across void, binder, and carbon fiber, confirming that mesoscale composition is learned and reproduced. For relative surface area (Fig. 6c), training and generated datasets follow the same statistical trends; a mild shift toward smoother binder surfaces is visible, consistent with convolutional generative models producing slightly regularized interfaces. Finally, the binder–carbon fiber relative interfacial area (Fig. 6d) is well preserved in distribution and mean, in line with the small bulk difference reported (~2.8%), indicating that the generator maintains physically meaningful phase boundaries and connectivity.

This trend reflects the inherent constraint of 2D convolutional layers in the GAN architecture, which lack full 3D spatial context, leading to more uniform connectivity patterns. In contrast, the relative surface area and interfacial area distributions of the real and generated datasets exhibit closely aligned trends with some minor tendency toward more smooth binder morphology in the generated dataset, which is consistent with the GAN's smoothing effect [27]. These trends of slight narrowing of variability align with what has been reported in previous studies by

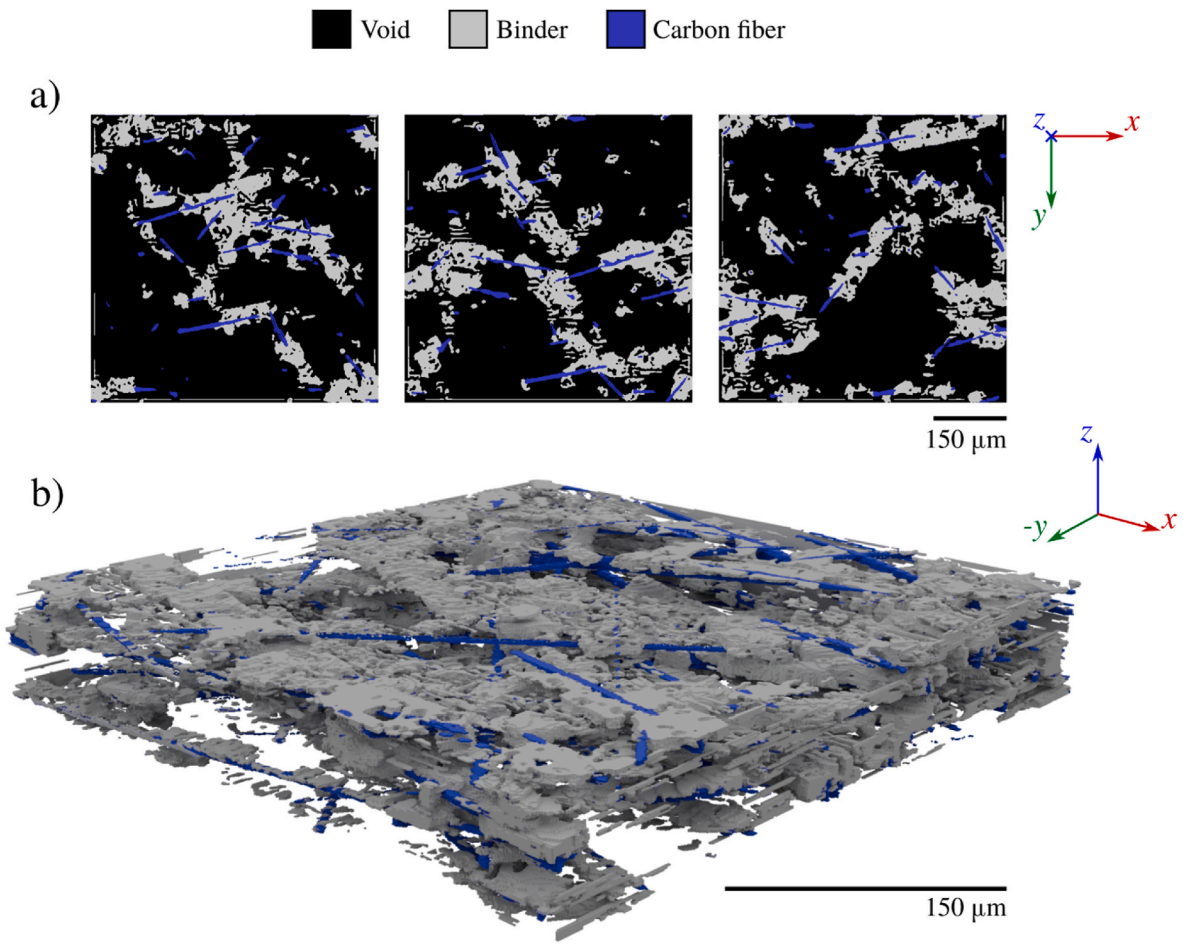


Fig. 4. Representation of the generated dataset. a) Generated 2D slices; b) 3D representation of the scans.

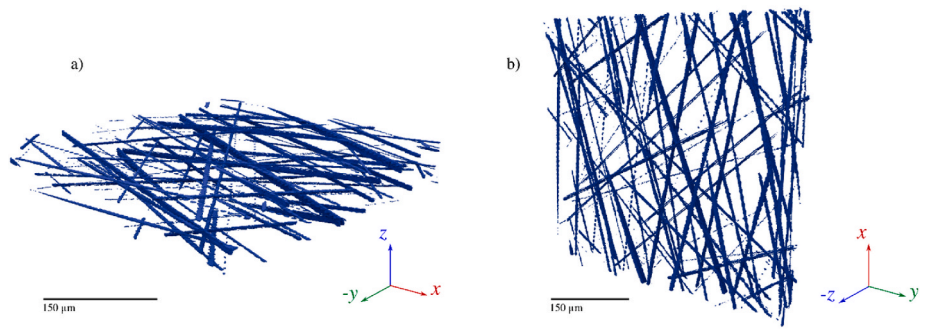


Fig. 5. Three-dimensional representation of carbon fibers in the generated structure. a) A view from the same perspective as in Fig. 4; b) A view from the bottom of the structure.

multiple authors [16,19,28]. Importantly, the model maintains the statistical trends and spatial heterogeneity essential for further electrochemical modelling.

While the generated microstructures accurately reproduce mean porosity and tortuosity, the smoothing effect inherent to convolutional architectures and the slight narrowing of tortuosity distributions represent a limitation, as they may bias transport predictions toward optimistic, non-conservative estimates in CFD or electrochemical simulations. This effect may be particularly relevant under high current density or mass-transport-limited operating conditions in PEM fuel cells, where local diffusion limitations govern polarization behavior. At the same time, a key strength of the generative approach is its ability to consistently capture the dominant geometric metrics that govern effective transport properties, ensuring reliable large-scale simulations with substantial computational efficiency compared to experimental reconstructions. Future work should focus on sensitivity analyses and validation against simulation-derived effective properties to assess the impact of metric deviations, alongside exploring architectural modifications that better preserve fine-scale heterogeneity i.e. multi-scale convolutional designs or hybrid generative frameworks that integrate physics-informed constraints.

These results demonstrate that the model reliably replicates global structural properties such as mean porosity and tortuosity, while partially capturing local features relevant to electrochemical performance. However, some fine-scale heterogeneity is smoothed, which may influence transport predictions under mass-transport-limited conditions.

Table 4
Bulk properties of the training dataset and one of the generated datasets. Relative surface area values are A_i/A_{void}. A_{void} is chosen as a reference because its value is usually the highest.

Property	Phase	Value (training dataset)	Value (generated dataset)
Tortuosity		1.9870	1.8528
Volume fraction	Void	73.41%	76.96%
	Binder	20.01%	20.05%
	Carbon fiber	3.75%	2.95%
	Other	2.83%	0.04%
Specific surface area, voxel^{-1}	Void	0.0849	0.0710
	Binder	0.761	0.0675
	Carbon fiber	0.0194	0.0182
Relative surface area	Void	1.000	1.000
	Binder	0.896	0.951
	Carbon fiber	0.228	0.256
Binder-carbon fiber interfacial area, voxel^{-1}		0.0072	0.0074
Binder-carbon fiber relative interfacial area		0.089	0.105

The ability to achieve this level of accuracy in under 35 min of computation highlights the efficiency and scalability of the C-GAN, positioning it as a viable alternative for digital twin applications and predictive modeling in hydrogen energy systems.

This work demonstrates that a conditional GAN can produce morphologically accurate, anisotropic 3D microstructures, especially with respect to layered anisotropy, which prior models underrepresented. The 2D convolutional generator, while computationally efficient, restricts long-range continuity along the z-axis. Although the hybrid kinematic simulation mitigates this by pre-aligning fibers, continuity artifacts can still emerge when fiber segments span >100 voxels.

The fiber - aligned preprocessing used in this study partially mitigates the limitations of 2D convolutional architectures by guiding the generator toward realistic anisotropic morphology. In future works, fully 3D GAN architectures could better address this limitation, however such models are computationally demanding. For example, training SliceGAN on subvolumes of 1283 voxels for 100 epochs required approximately 9.5 h and 13 GB of VRAM on our GPU. These subvolumes fail to capture the complete material structure, and even the largest domain used in our experiments (2563 voxels) was insufficient to preserve global features. These discontinuities affect fiber alignment and may degrade transport simulations.

4. Conclusion

In this work, we develop and validate a conditional generative adversarial network (C-GAN) framework for reconstructing three-dimensional, anisotropic, multiphase microstructures in PEM fuel cell gas diffusion layers and demonstrate its ability to accurately capture key structural and transport-relevant characteristics. The main conclusions of this study are as follows.

(i) The proposed framework successfully reconstructs anisotropic, multiphase electrode structures while preserving key geometrical and transport-relevant metrics, including tortuosity, interfacial area, phase distribution, and anisotropic fiber alignment, within acceptable deviations.
(ii) The generated microstructures capture both local features and global morphological characteristics, such as elongated fiber domains, interfacial boundaries, and layered anisotropy, consistent with real PEMFC GDL architectures across multiple planes.
(iii) The framework enables efficient large-scale generation of volumetric datasets (approximately 100 vol in under 35 min), establishing GAN-based reconstruction as a scalable, computationally efficient, and physics-aware alternative to data-intensive imaging

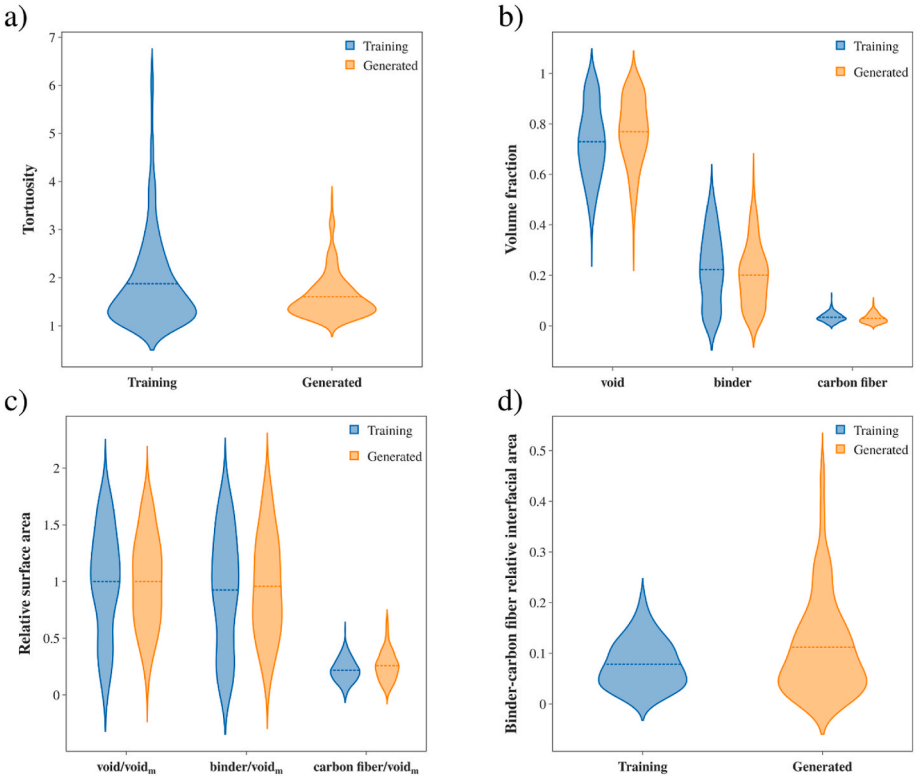


Fig. 6. Distributions of physical metrics using samples of 300 random subvolumes of dimensions 100 × 100 × 25 voxels vs: a) Tortuosity; b) Phase volume fractions (void, binder, carbon fiber); c) Relative surface area (normalized to the void phase); d) Binder–carbon fiber relative interfacial area.

methods for digital twin applications and predictive structure-property analysis in hydrogen energy systems.

Future work will focus on implementing fully three-dimensional generative architectures enabled by high-performance computing resources. In particular, a 3D attention module will be introduced to explicitly enforce long-range structural coherence by penalizing discontinuities in fiber trajectories exceeding a predefined voxel threshold (e.g., >100 voxels). In parallel, a composite physics-informed loss function can be developed, directly incorporating transport-relevant descriptors, i.e. TauFactor - calculated tortuosity and phase percolation metrics as regularization terms during training. This approach will enable quantitative coupling between generative reconstruction and transport-relevant morphological constraints. Additionally, hybrid GAN - diffusion architectures could be explored to enhance texture fidelity and structural continuity across extended length scales.

CRediT authorship contribution statement

Daulet Tazhbagambetov: Writing – original draft, Visualization, Software, Methodology. **Azamat Zhaksylykov:** Writing – original draft, Visualization, Methodology, Formal analysis. **Aliya Toleuova:** Validation, Investigation, Conceptualization. **Iliyas Tursynbek:** Validation, Supervision, Methodology. **Diar Demeubay:** Software, Methodology, Investigation. **Kunning Tang:** Resources, Methodology, Data curation. **Chuan Zhao:** Writing – review & editing. **Quentin Meyer:** Writing – review & editing, Data curation. **Assiya Yermukhambetova:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijhydene.2026.154671.

Data availability

The 3D datasets presented in this study are available from the corresponding author on reasonable request.

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